



Service Tool Instruction

Service Tool Instruction
Number: 3377927

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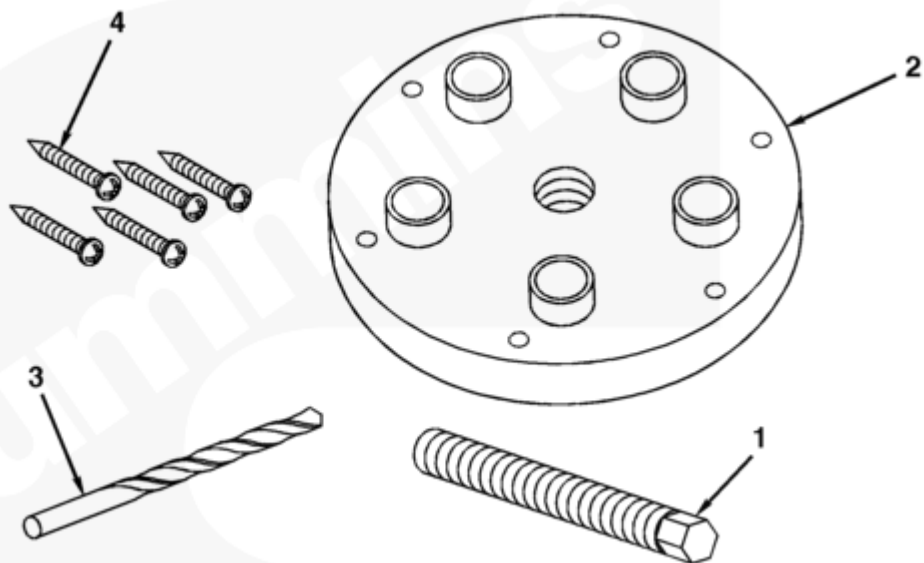
Description

Rear Crankshaft Seal Replacer Kit

Purpose

This service tool instruction explains the use of the rear crankshaft seal replacer kit, Part Number 3164927, for the QSK23 engine.

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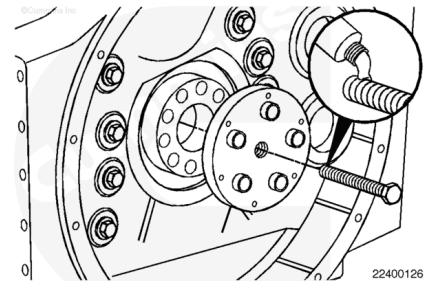
Table 1, Rear Crankshaft Seal Replacer Kit, Part Number 3164927

Item Number	Part Number	Description	Quantity
1	3164967	Replacer screw	1
2	3164966	Crankshaft seal replacer	1
3	3164218	Drill, [9/64-inch]	1
4	3164217	Sheet metal screw, number 10 x 25.4- mm [1-in] (5 shown)	36

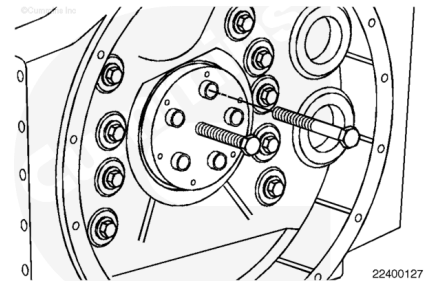
Lubricate the threads and tip of the replacer screw with Lubriplate™, Part Number 3163086, Part Number 3163087 or a suitable grease.

Install the replacer screw into the crankshaft seal replacer. Make sure the replacer screw tip is flush with the inside of the crankshaft seal replacer.

Position the crankshaft seal replacer against the end of the crankshaft.



Install one flywheel mounting capscrew through the crankshaft seal replacer into the crankshaft and finger-tighten. This will hold the replacer in position until the sheet metal screws are installed.

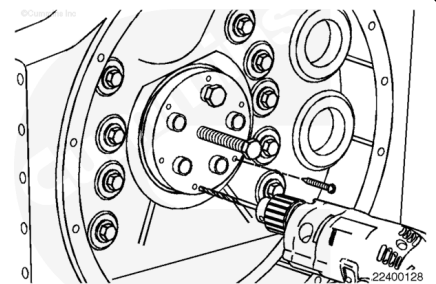


⚠ CAUTION ⚠

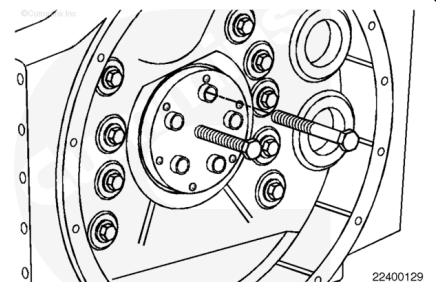
Drill holes carefully and straight to reduce the possibility of damage to the flywheel housing or the crankshaft.

Apply grease to the drill to catch the chips. Stop frequently to remove the chips.

Drill one hole 10 mm [0.39 in] deep through the oil seal casing. Install the sheet metal screw into the hole. Drill five additional holes 10 mm [0.39 in] deep and install the sheet metal screws. Make sure all six sheet metal screws are threaded through the oil seal casing and uniformly positioned against the crankshaft seal replacer.



Remove the flywheel mounting capscrew.



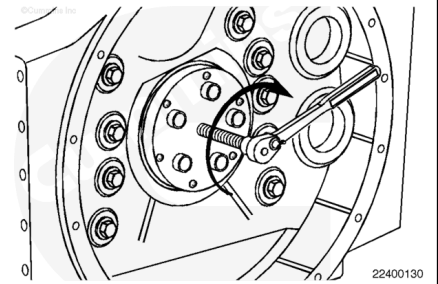
⚠ CAUTION ⚠

Do not use an impact wrench or air tools. Doing so can damage the tool.

Slowly rotate the replacer screw **clockwise** until the oil seal is removed.

If the sheet metal screws pull out of the seal, stop the removal operation. Remove the sheet metal screws, reorient the crankshaft seal replacer, drill new holes, and reinstall the sheet metal screws.

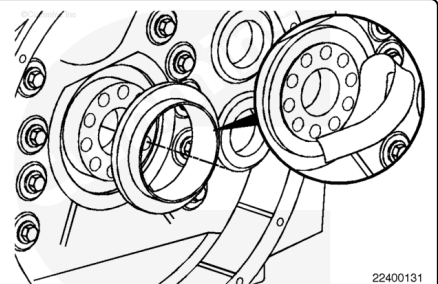
Complete the removal procedure, remove the replacer, and discard the oil seal.



⚠ CAUTION ⚠

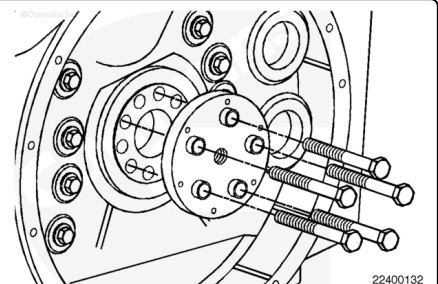
Before installing the oil seal make sure all the metal chips are completely removed and the surface is clean. Failure to do so can cause engine damage.

Using the plastic installation sleeve, place the new oil seal over the rear crankshaft nose and slide the oil seal toward the flywheel housing. Make sure the oil seal is positioned squarely on the crankshaft. Avoid cocking the seal. Remove the plastic installation sleeve supplied with the oil seal.



Remove the replacer screw from the crankshaft seal replacer. Position the replacer over the crankshaft and against the oil seal.

Install five flywheel mounting capscrews through the replacer into the crankshaft and uniformly finger-tighten.



Tighten the first capscrew 1/4 of a turn.

Finger-tighten the remaining capscrews. After the capscrews are finger-tight, tighten the capscrews 1/4 of a turn in an every other capscrew pattern.

Continue tightening the flywheel mounting capscrews until the crankshaft seal replacer is positioned against the crankshaft.

Remove the flywheel mounting capscrews and the replacer.

